

TITLE

AUTONOMOUS MOBILE ROBOT (AMR)
NAVIGATION USING A* PATH PLANNING

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ABSTRACT

Autonomous Mobile Robots (AMRs) are intelligent robotic systems capable of navigating an environment without human intervention. The objective of this project is to simulate an AMR capable of mapping its surroundings, finding the shortest path between two points, and dynamically avoiding obstacles. The project uses Occupancy Grid Mapping, A* Path Planning Algorithm, and Dynamic Obstacle Avoidance techniques implemented in Python.

OBJECTIVE

The main objectives of this project are:

1. To create a 2D occupancy grid map of the environment.
2. To implement the A* path planning algorithm.
3. To determine the shortest route between start and goal locations.
4. To perform obstacle detection and path replanning.
5. To simulate autonomous navigation behaviour.

SOFTWARE AND TOOLS USED

- . Python
- . NumPy
- . Matplotlib
- . Google Colab
- . GitHub

INTRODUCTION

Autonomous Mobile Robots are widely used in warehouses, manufacturing industries, hospitals, and smart transportation systems. For successful navigation, a robot must understand its environment, determine an efficient path, and react to unexpected obstacles.

This project demonstrates these capabilities through a simulated environment represented by an occupancy grid. The A* algorithm is used to compute the shortest path while avoiding obstacles.

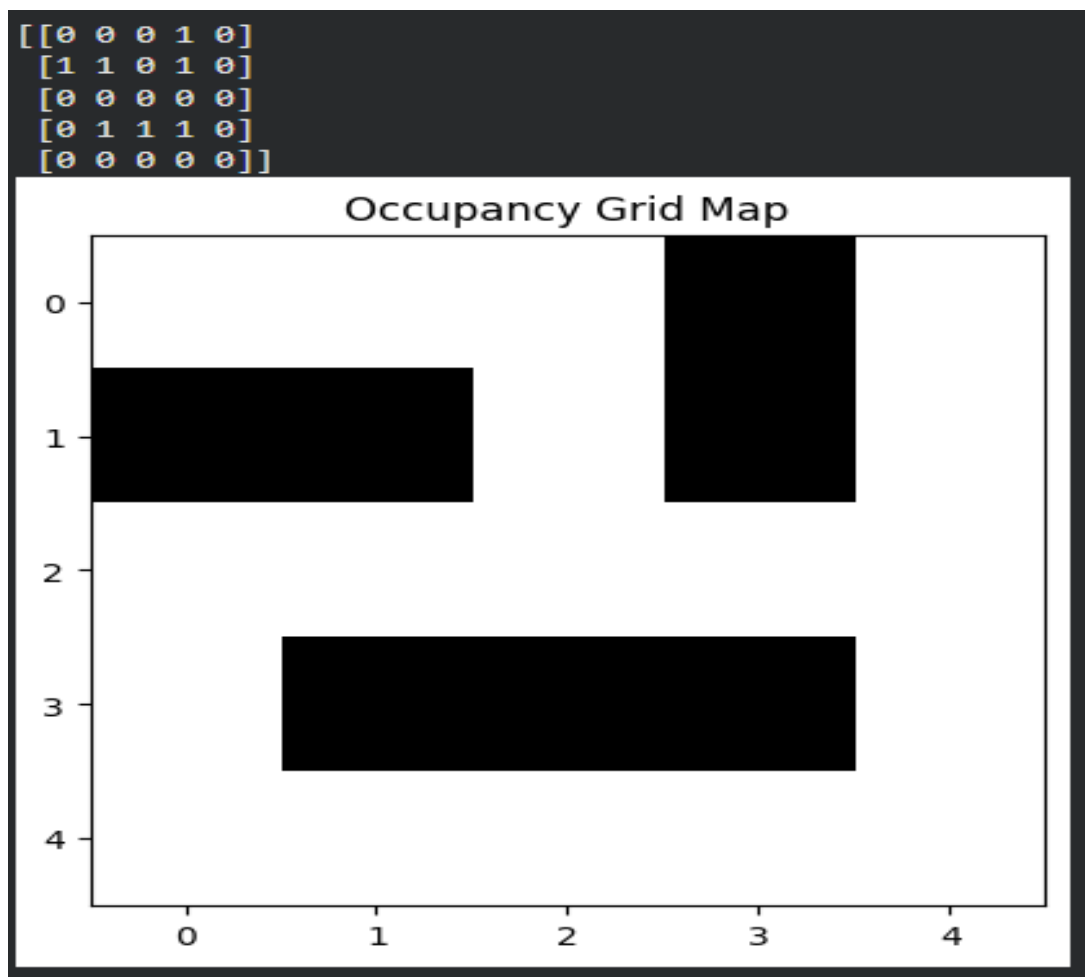
METHODOLOGY

Step 1: Occupancy Grid Mapping

The environment is represented as a grid where:

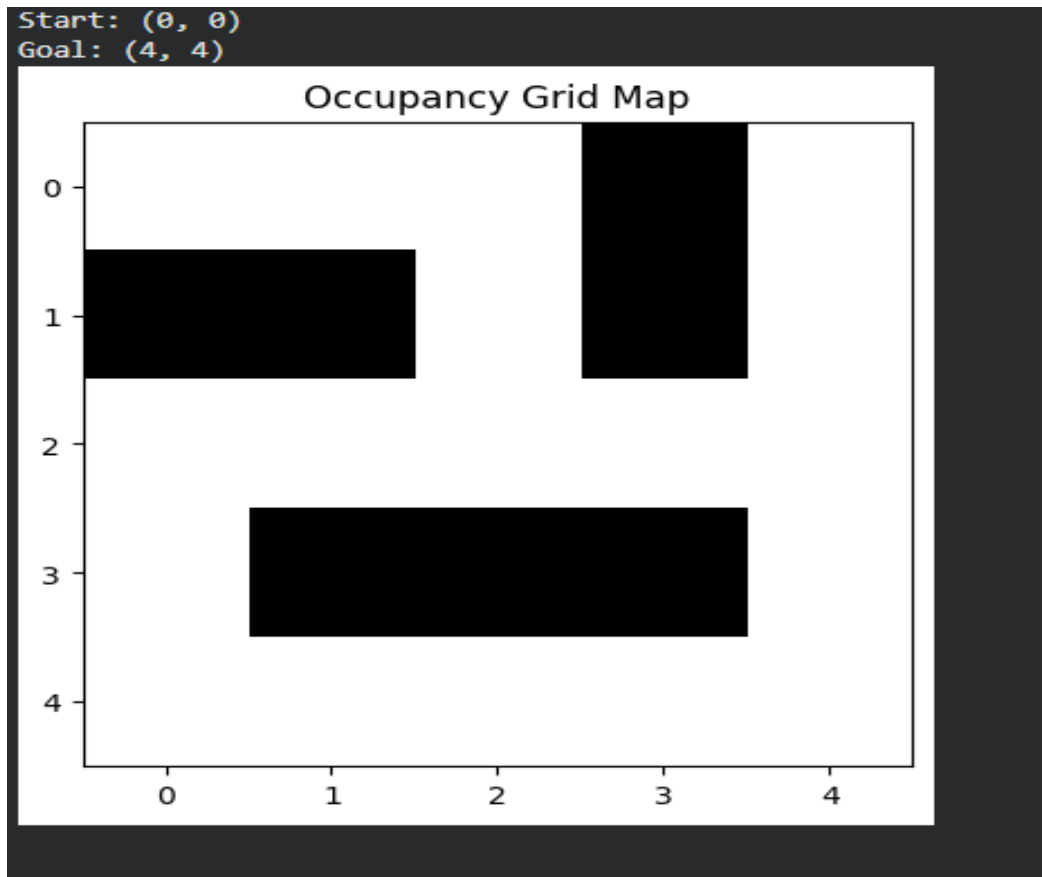
- . 0 = Free Space
- . 1 = Obstacle

The robot uses this grid to understand navigable and blocked regions.



Step 2: Defining Start and Goal Positions

The robot starts from coordinate (0,0) and aims to reach coordinate (4,4).

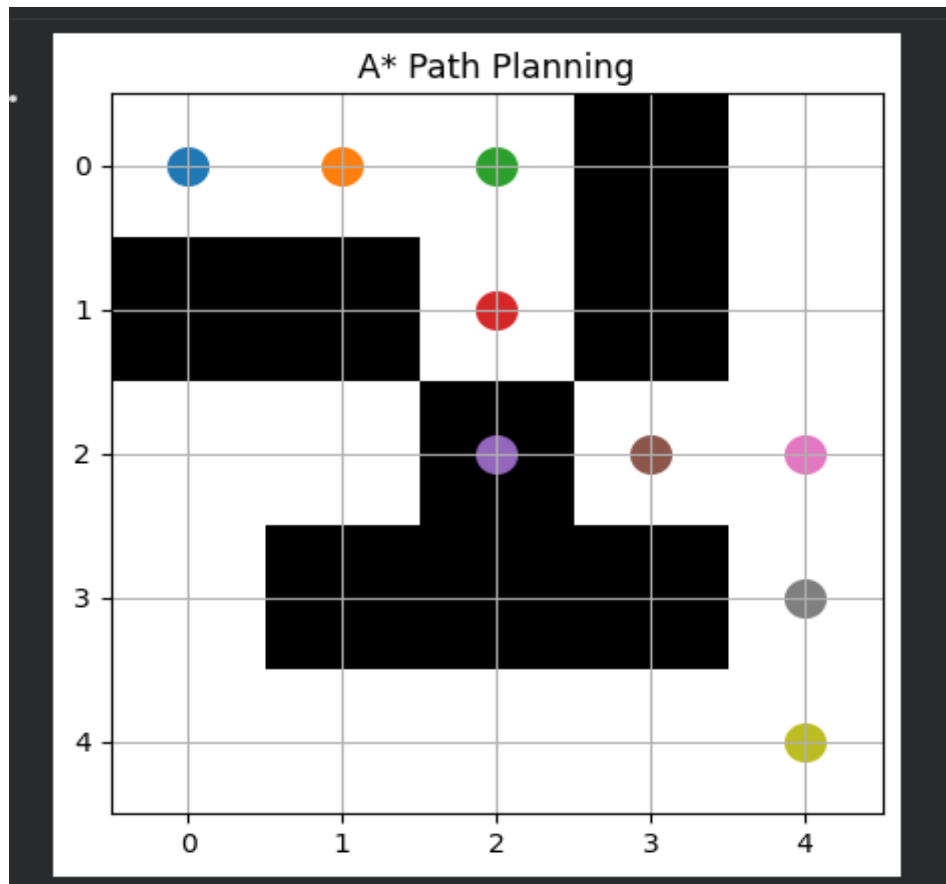


Step 3: A* Path Planning

The A* algorithm calculates the shortest possible path from the start point to the destination while avoiding obstacles.

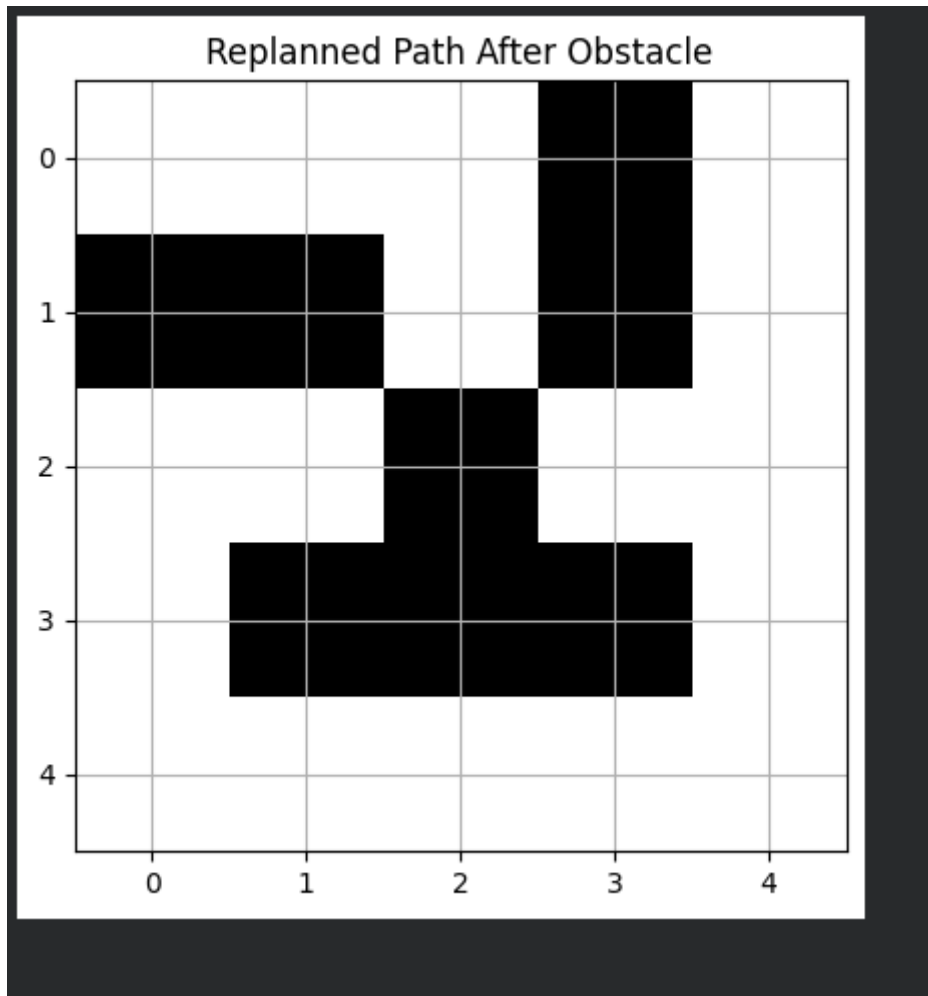
Advantages of A* Algorithm:

- . Fast execution
- . Optimal path generation
- . Efficient obstacle avoidance



Step 4: Dynamic Obstacle Avoidance

A new obstacle is introduced into the environment during execution. The robot detects the change and recalculates an alternative route.



RESULTS

The developed system successfully:

- . Created an occupancy grid map.
- . Planned the shortest route using the A* algorithm.
- . Detected newly introduced obstacles.
- . Replanned the path automatically.
- . Demonstrated autonomous navigation behaviour.

ADVANTAGES

- . Efficient path planning.
- . Reduced travel distance.
- . Dynamic obstacle handling.
- . Improved navigation accuracy.
- . Scalable for larger robotic systems.

APPLICATIONS

- . Warehouse Automation
- . Industrial Robotics
- . Delivery Robots
- . Smart Transportation Systems
- . Autonomous Vehicles
- . Hospital Service Robots

CONCLUSION

The Autonomous Mobile Robot Navigation project was successfully implemented using Python. The system demonstrated occupancy grid mapping, shortest path calculation using the A* algorithm, and dynamic obstacle avoidance through path replanning. The project successfully satisfies the requirements of autonomous robot navigation and provides a foundation for advanced robotic applications.

REFERENCES

1. Python Documentation
2. NumPy Documentation
3. Matplotlib Documentation
4. A* Path Planning Algorithm Resources
5. DecodeLabs Robotics & Automation Internship Material